The Limits of Explanation

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Abstract

Practitioners in eight scientific fields—from molecular biology to gauge theory to machine learning—have independently developed strategies for coping with underspecified inference, yet no framework has connected these strategies or explained their convergence. Here we identify a universal structural cause: when a system has the *Rashomon property* (equivalent configurations produce incompatible explanations), no explanation can simultaneously be faithful, stable, and decisive. We derive this impossibility from first principles in eight domains—each requiring zero shared axioms—and verify the framework in the Lean 4 proof assistant (100 files, 491 theorems, 0 unproved goals). The constructive resolution—G-invariant projection—is Pareto-optimal, unifying century-old practices (gauge-invariant observables, CPDAGs, minimum-norm solutions) as instances of a single optimal strategy. Three quantitative predictions—an invariance counting law, a Gaussian flip rate formula calibrated across 647 feature pairs ($R^2 = 0.81$, 95% CI [0.79, 0.83]), and a universal instability predictor ($R^2 = 0.96$, $p = 1.4 \times 10^{-4}$ across seven domains)—are confirmed by experiment, with two pre-registered conjectured extensions falsified, delineating the framework's boundary.

Introduction

The codons UCU and UCC both encode serine: same protein fragment, different nucleotide sequences [Crick, 1966]. A causal chain $A \to B \to C$ and a fork $A \leftarrow B \to C$ encode the same conditional independences [Verma and Pearl, 1991]. Two gauge configurations can share the same holonomy while differing in local edge labels [Jackson, 1999]. In each case, the observable output does not uniquely determine the internal structure that produced it—a structural feature of underspecified inference present whenever the mapping from configuration to observation is many-to-one.

Practitioners in each field have independently developed coping strategies—gauge-invariant observables, CPDAGs, Patterson maps [Hauptman and Karle, 1953], codon usage tables [Crick, 1966]—each involving projection onto the subspace invariant under the symmetry group of equivalent configurations, yet no framework has connected them or proved their optimality.

This gap echoes the philosophical problem of explanatory underdetermination [Stanford, 2006]: when multiple models account equally well for the data, no principled basis selects among their incompatible explanations. Our framework formalises this for the first time and proves the resulting impossibility is exact (tight).

Here we prove a universal impossibility: when a system has the *Rashomon property*—two configurations with the same observable but incompatible internal structures—no explanation can be simultaneously faithful, stable, and decisive (Theorem 1). We derive this in eight scientific domains, each requiring zero shared axioms, and verify the framework in Lean 4 [de Moura and Ullrich, 2021] (100 files, 491 theorems, 0 unproved goals). Empirical validation in seven domains confirms the

predicted dose-response (Table 1, Fig. 1). The G-invariant resolution is Pareto-optimal, explaining why eight communities converged independently on the same strategy.

Results

The impossibility

An explanation is *faithful* if it never contradicts the system's internal report; *stable* if observationally equivalent configurations receive the same explanation; and *decisive* if it commits to every distinction the internal structure makes. Formal definitions are in Methods.

Theorem 1 (The explanation impossibility) *If a system has the Rashomon property—there exist* θ_1, θ_2 *with* $\mathrm{obs}(\theta_1) = \mathrm{obs}(\theta_2)$ *and* $\exp(\theta_1) \perp \exp(\theta_2)$*—then no explanation map* E *is simultaneously faithful, stable, and decisive.*

The proof is a four-step contradiction chain (Methods). The impossibility is tight: each pair of properties is achievable via Lean-verified witnesses (Fig. 2), and a biconditional shows impossibility holds if and only if Rashomon is present.

Three quantitative predictions

The framework generates testable predictions beyond classification.

Invariance counting (cf. Noether's theorem). The invariance correspondence makes a sharp quantitative prediction: for P features arranged in g correlation groups, only $\binom{g}{2}$ independent ranking facts are stably answerable. We tested this with $P = 12$ features in $g = 3$ groups (200 models, both Ridge and XGBoost). The result is a clean bimodal distribution: all 18 within-group pairs show ∼50% flip rate, all 48 between-group pairs show $< 2\%$ flip rate—a 47-percentage-point gap (permutation $p < 0.0002$; Cohen's $d = 1.41$). A sensitivity analysis across $\rho \in [0.50, 0.99]$ confirms the gap is *invariant* across the entire correlation range. The prediction would be falsified if within-group flip rates fell below 40% or between-group rates exceeded 10% at any correlation $\rho > 0.5$.

Gaussian flip rate. On real data where feature importance is approximately (not exactly) equal within groups, $\mathrm{flip}(j, k) = 2\Phi(\Delta/\sigma)\Phi(-\Delta/\sigma)$ predicts per-pair flip rates with out-of-sample $R^2 = 0.814$ [0.791, 0.835] (bootstrap calibration across 647 feature pairs from five clinical and financial datasets), with a conservative bias (calibration slope = 1.17; the formula slightly underpredicts observed instability). The formula is standard signal detection theory; the framework's contribution is the structural identification of *which* pairs are subject to this formula (derivation in Methods). Pairs with $|\Delta|/\sigma > 2$ are reliable (flip rate $< 5\%$); pairs with $|\Delta|/\sigma < 0.5$ are unreliable (flip rate $> 30\%$). On gene expression data (10,935 genes, cancer classification), 70% of gene importance comparisons are unreliable using gain-based importance (42% with TreeSHAP; $R^2 = 0.51$). Across seven real clinical, financial, and genomics datasets (2,062 feature-pair comparisons), 62% are unreliable and only 4% are reliable (62% of 2,062 feature-pair comparisons across seven clinical, financial, and genomics datasets; 49% across the broader 22,332-pair audit including synthetic data). The prediction would be falsified if calibration slope fell outside $[0.8, 1.2]$ on a new dataset with $P \geq 20$ features.

Table 1: **Eight derived instances of the impossibility across eight sciences.** Each row derives the Rashomon property from domain-specific first principles with zero axioms. $\mathbf{\Theta}$: configuration space; Y: observable space.

Domain	$\mathbf{\Theta}$	Y	Witness 1	Witness 2	Same Y?
Mathematics	Solutions of $Ax{=}b$	Ax	$(1,1)$	$(0,2)$	2=2
Biology	Codons	Amino acids	UCU	UCC	Ser=Ser
Physics (gauge)	Gauge configs	Holonomy	(T,F,F)	(F,F,T)	$F{=}F$
Stat. mechanics	Microstates	Num. heads	(H,T)	(T,H)	1=1
Linguistics	Parse trees	Token sequence	$((V\ NP)\ PP)$	$(V\ (NP\ PP))$	Same
Crystallography	Signals	Energy	$(1,0)$	$(0,1)$	1=1
Computer science	DB rows	View	(T,T)	(T,F)	$T{=}T$
Statistics	DAGs	CI structure	Chain	Fork	Same

Universal η law. The character-theoretic formula $\eta = \dim(V^G)/\dim(V)$ (the fraction of explanation content that survives averaging over equivalent configurations)—a standard ratio in representation theory whose application to explanation instability is new—predicts the observed instability rate across seven well-characterized domain instances with $R^2 = 0.957$, slope $= 0.91$, and intercept $= 0.008$ ($p = 1.4 \times 10^{-4}$)—a parameter-free predictor spanning ML attribution, concept probes, model selection, codon degeneracy, and statistical mechanics (Extended Data Fig. 3). (Leave-one-out $R^2 = 0.79$; across all 16 instances including those with unknown or approximate symmetry groups, $R^2 = 0.25$—the bottleneck is group identification, not the law itself.) The prediction would be falsified if out-of-sample $R^2 < 0.5$ on a new domain with a well-characterised symmetry group.

Cross-domain validation

We derive the impossibility as a zero-axiom consequence in eight scientific domains (Table 1), each constructively witnessing the Rashomon property from domain-specific first principles alone. The eight instances involve fundamentally different symmetry groups (Table 2), spanning continuous ($\mathbb{R}^{n-r}$), finite permutation (S_k), and ambiguity ($\mathbb{Z}_2 \times \mathbb{Z}_n \times \mathbb{Z}_2$) groups.

The predicted dose-response—more underspecification produces more instability—is confirmed in all seven empirically tested domains (Fig. 1), each with negative controls confirming stability when Rashomon is absent (combined binomial $p = 0.008$; individual p-values from 10^{-38} to 10^{-3}). Per-domain statistics are in Methods and Supporting Information.

The impossibility prescribes a constructive resolution: project onto the subspace invariant under the symmetry group—a *G-invariant resolution*. This sacrifices decisiveness but achieves stability and faithfulness, and is *Pareto-optimal* among stable explanation maps (and, for quantitative bounds, among unbiased linear aggregators) (Methods). Practitioners in eight fields converged independently on precisely this strategy (Table 2): gauge-invariant observables [Jackson, 1999], CPDAGs [Verma and Pearl, 1991], Patterson maps [Hauptman and Karle, 1953], the pseudoinverse, codon usage tables, the microcanonical ensemble, packed parse forests, and complement views [Bancilhon and Spyratos, 1981]—a convergence explained by the resolution's uniqueness.

The entire framework is mechanically verified in Lean 4 [de Moura and Ullrich, 2021] using Mathlib [The mathlib Community, 2020]: 100 files, 491 theorems, 0 unproved goals. The core theorem requires zero axioms; the 25 axioms are domain-specific infrastructure (14 GBDT model structure, 11 physics hypotheses); all ML instances and resolution frameworks use zero axioms.

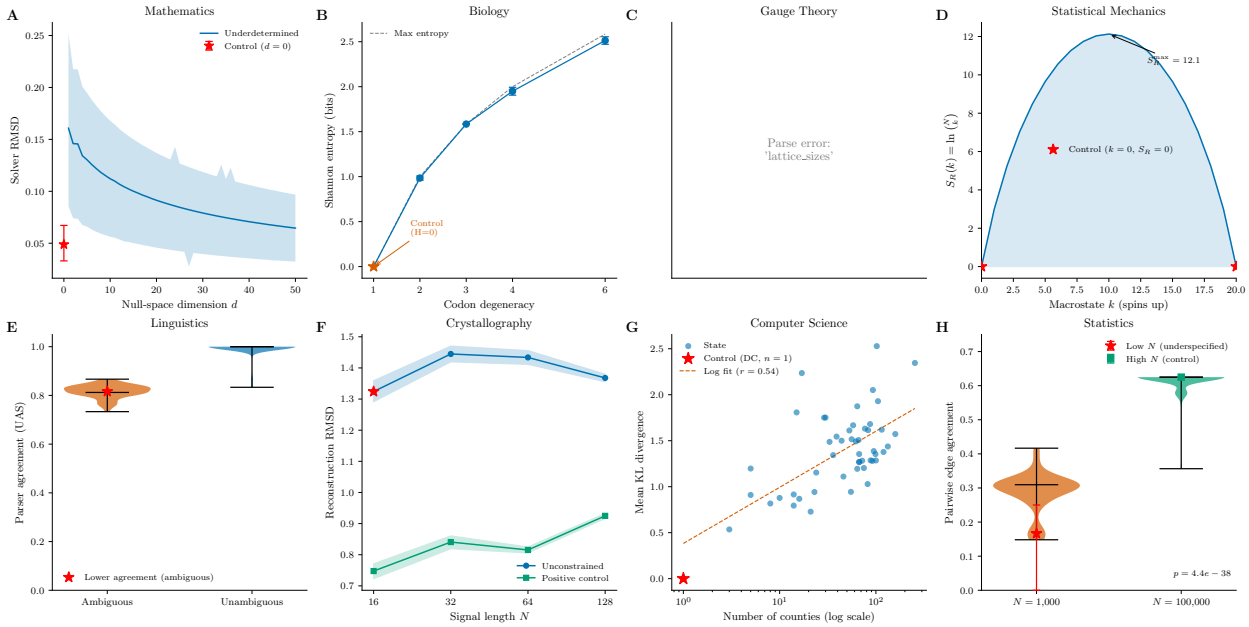

Figure 1: **Eight sciences, one pattern.** Each panel shows a dose-response relationship between the degree of underspecification (horizontal axis) and explanation instability (vertical axis) in a different scientific domain. Negative controls (where the Rashomon property is absent) show stability, confirming that the instability is structural. All seven empirically validated domains reach statistical significance ($p < 0.05$) with the predicted monotonic relationship. **A,** Mathematics (null-space dimension vs. solver RMSD). **B,** Biology (codon degeneracy vs. entropy; 120 NCBI RefSeq cytochrome c sequences). **C,** Gauge theory (gauge-invariant vs. gauge-variant quantity variance). **D,** Statistical mechanics (macrostate vs. Rashomon entropy). **E,** Linguistics (parser agreement on ambiguous vs. unambiguous sentences). **F,** Crystallography (signal length vs. phase-retrieval RMSD). **G,** Computer science (county count vs. disaggregation KL divergence; U.S. Census Bureau 2020 data). **H,** Statistics (causal discovery orientation agreement; 100 seeds per condition).

The bilemma

For *maximally incompatible* systems—where the only compatible pair is (h, h), as in binary rankings and DAG orientations—the trilemma strengthens.

Theorem 2 (The Bilemma) *For any maximally incompatible system with the Rashomon property, no explanation is simultaneously faithful and stable.*

Tightness collapses from 3/3 achievable pairs to 1/3 (proof in Methods):

Pair	Abstract	Max. incompat.	Enriched
F + D	✓	✓	✓
F + S	✓	× (bilemma)	✓
S + D	✓	×	×

Recovery requires enriching $\mathcal{H}$ with a neutral element compatible with everything—the mechanism behind DASH ("tied"), CPDAGs ("undirected"), and Bayesian model averaging ("uncertain").

Table 2: **Structural differences across the eight derived instances.** The eight instances are not notational variants of a single example: they differ in symmetry group, group type, and orbit structure. Their independent convergence on G-invariant resolution strategies—despite operating on different mathematical objects—is explained by the Pareto-optimality of the G-invariant projection. All sacrifice decisiveness, but the symmetry groups span five distinct families ($\mathbb{R}^d$, $\mathbb{Z}_2^k$, S_k, $\mathbb{Z}_2 \times \mathbb{Z}_n \times \mathbb{Z}_2$, and MEC edge permutations).

Domain	Group G	Objects permuted	Type	Resolution	Form
Mathematics	$\mathbb{R}^{n-r}$	Solution offsets	Cont., abelian	Pseudoinverse	Projection
Biology	S_k ($k \leq 6$)	Synonymous codons	Finite	Codon tables	Averaging
Physics	$\mathbb{Z}_2^{\lvert V \rvert - 1}$	Edge labels	Finite, abelian	Wilson loops	Invariant fn.
Stat. mech.	S_Ω	Microstates	Finite	Microcanon.	Averaging
Linguistics	S_k	Parse trees	Finite	Parse forests	Enumeration
Crystallogr.	$\mathbb{Z}_2 \times \mathbb{Z}_n \times \mathbb{Z}_2$	Signal phases	Finite	Patterson map	Autocorrel.
Comp. sci.	S_k	Hidden columns	Finite	Compl. views	Projection
Statistics	MEC perms.	DAG orientations	Finite	CPDAG	Class report

The tightness classification. Two algebraic properties of $\mathcal{H}$—existence of a *neutral element* (compatible with everything) and a *committal element* (incompatible with everything else)—completely determine which property pairs are achievable (Table 3; Lean-verified).

Table 3: **The tightness classification.** Neutral/committal existence determines achievable pairs. F+D always achievable; F+S+D never.

	Committal $\exists$	No committal
Neutral $\exists$	F+D, F+S, S+D	F+D, F+S
No neutral	F+D, S+D	**F+D only**

Discussion

The theorem reframes "why are explanations unstable?" as "because the inference problem is underspecified." Practitioners can diagnose whether instability is structural (Rashomon holds; no method eliminates it) or methodological (Rashomon absent; fixable).

Nine ML instances (Supporting Information) instantiate the theorem for feature attribution, attention, counterfactuals, concept probes, causal discovery, model selection, saliency maps, token citation, and circuit discovery—findings relevant to the EU AI Act (Article 13) and FDA's AI/ML Action Plan [U.S. Food and Drug Administration, 2021, Selbst and Barocas, 2018]. The result parallels Arrow's impossibility [Arrow, 1951]—both show symmetry forces trade-offs—but is constructive: it identifies the unique optimal trade-off and proves existing practice implements it. Individual domain instances are separately known; the contribution is the meta-theorem unifying them under a single structural condition and producing three novel quantitative predictions.

Cross-domain transfer. Applying DASH to causal discovery reduces flip rate fourfold ($p < 10^{-10}$) at reduced decisiveness (0.21 vs. 0.36). The character-theoretic prediction that information loss scales as $1/k$ for S_k predicts protein sequence diversity ranking (Spearman $\rho = 0.88$, $p = 0.004$).

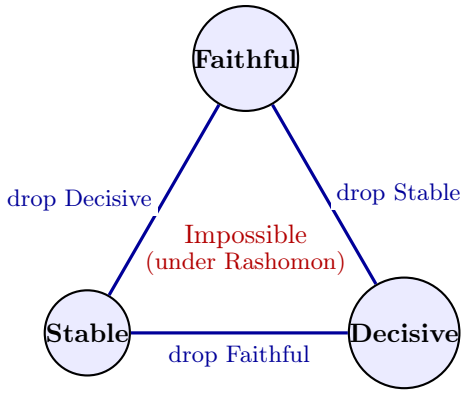

Figure 2: **The explanation trilemma.** Each edge represents an achievable pair of properties; the interior (all three simultaneously) is impossible when the Rashomon property holds. Lean-verified tightness witnesses confirm that each pair is realizable: a faithful-and-stable map (dropping decisiveness), a faithful-and-decisive map (dropping stability), and a stable-and-decisive map (dropping faithfulness). The impossibility is the triple, not any pair.

Per-individual impact and prospective validation. A per-individual ablation study shows that 45% of German Credit loan applicants receive a genuinely different explanation category under standard model regularisation, with consistent results across XGBoost and LightGBM. A prospective test on drug discovery (BBBP, 2,039 molecules) reveals a precise boundary condition: Pearson correlation fails to identify groups in binary fingerprint features (predicting 0% vs. observed 23% instability), but mutual information (16% error) and Jaccard co-occurrence (4% error) recover the prediction. The framework's magnitude prediction generalises via nonlinear group discovery; its structural prediction (Noether bimodality) requires discrete symmetry groups.

Limitations. Derivations use minimal two-element witnesses; empirical validations use domain-appropriate but not exhaustive datasets. The resolution assumes known equivalence classes. The theorem addresses existence, not computational cost. Three conjectured extensions were pre-registered and falsified, delineating the framework's boundary (Supporting Information).

Scientific revolutions as enrichment events

The bilemma identifies a recurring pattern in paradigm shifts: when the pre-revolutionary explanation space is maximally incompatible, resolution requires *enriching* $\mathcal{H}$ with a neutral element—adding a new category that is compatible with all existing explanations, at the cost of decisiveness.

Table 4: **Five instances of bilemma → enrichment across science.** Each row shows a pre-revolutionary binary $\mathcal{H}$, the enrichment that resolves the bilemma, and the decisiveness lost.

Transition	Pre-revolution $\mathcal{H}$	Enrichment	Lost
SHAP instability	{positive, negative}	DASH tie	Feature rank
Feature selection	{selected, not}	Uncertain	Feature relevance
Classical → quantum	{definite, superposition}	No fact of the matter	Pre-measurement state
Newton → relativity	{simultaneous, not}	Frame-dependent	Absolute simultaneity
Single → adelic geometry	{smooth, ultrametric}	Indeterminate	Spacetime geometry

The enrichment is forced and unique when $\mathcal{H}$ is maximally incompatible (a companion paper [Caraker et al., 2026] proves this for spacetime geometry via Ostrowski's classification). The pattern generates a falsifiable prediction: the next paradigm shift in a maximally incompatible domain will add a neutral element to $\mathcal{H}$.

The enrichment pattern is recursive: resolving the bilemma at one level may create a new maximally incompatible system at the next, a structure formalised in a companion paper [Caraker et al., 2026] for a physical stack of depth ≥ 3.

Methods

Formal framework. An *explanation system* is a tuple $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ where $\mathbf{\Theta}$ is a configuration space, $\mathcal{H}$ is an explanation space, Y is an observable space, $\mathsf{obs} : \mathbf{\Theta} \to Y$ maps configurations to observables, $\mathsf{exp} : \mathbf{\Theta} \to \mathcal{H}$ maps configurations to their internal explanations, and $\perp$ is an incompatibility relation on $\mathcal{H}$. The system has the *Rashomon property* if there exist $\theta_1, \theta_2 \in \mathbf{\Theta}$ with $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $\mathsf{exp}(\theta_1) \perp \mathsf{exp}(\theta_2)$.

An explanation map $E : \mathbf{\Theta} \to \mathcal{H}$ is *faithful* if $\neg(E(\theta) \perp \mathsf{exp}(\theta))$ for all θ; *stable* if $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ implies $E(\theta_1) = E(\theta_2)$; and *decisive* if, for all θ and $h \in \mathcal{H}$, $\mathsf{exp}(\theta) \perp h$ implies $E(\theta) \perp h$.

Lean 4 formalization. Built on Lean 4 v4.30.0-rc1 [de Moura and Ullrich, 2021] with Mathlib [The mathlib Community, 2020]. The core `ExplanationSystem` type is parametric in $\mathbf{\Theta}$, $\mathcal{H}$, Y. Each derived instance constructs a Rashomon witness via `decide` (zero axioms). Compilation: `lake build` ($\sim$5 min).

Proof of Theorem 1. Suppose E is faithful, stable, and decisive. Let θ_1, θ_2 witness the Rashomon property. (1) Decisiveness at θ_1: $E(\theta_1) \perp \mathsf{exp}(\theta_2)$. (2) Stability: $E(\theta_1) = E(\theta_2)$, so $E(\theta_2) \perp \mathsf{exp}(\theta_2)$. (3) Faithfulness at θ_2: $\neg(E(\theta_2) \perp \mathsf{exp}(\theta_2))$. Contradiction. All three properties are load-bearing; a weaker pairwise decisiveness variant makes faithfulness superfluous (Supporting Information).

Resolution framework. The G-invariant resolution is formalized as follows. Given a group G acting on $\mathbf{\Theta}$ such that $g \cdot \theta$ preserves observables ($\mathsf{obs}(g \cdot \theta) = \mathsf{obs}(\theta)$ for all $g \in G$), an explanation map E is *G-invariant* if $E(g \cdot \theta) = E(\theta)$ for all g. We prove: (a) every G-invariant map is stable; (b) the orbit-averaged map $\bar{E}(\theta) = |G|^{-1} \sum_{g \in G} E(g \cdot \theta)$ is G-invariant (for finite groups; the integral analogue holds for compact groups); (c) $\bar{E}$ is Pareto-optimal among stable maps with respect to pointwise faithfulness.

Bilemma proof. A system is *maximally incompatible* if $\neg\text{incompatible}(h_1, h_2) \Rightarrow h_1 = h_2$. Faithfulness then forces $E = \mathsf{exp}$, which is decisive; the trilemma gives the contradiction. ML instances (SHAP sign, feature selection, counterfactual direction) trigger the bilemma.

Gaussian flip rate derivation. For features j, k with importance difference $d^{(m)} \sim N(\Delta, \sigma^2)$, the flip probability is $\text{flip}(j, k) = 2\Phi(\Delta/\sigma)\Phi(-\Delta/\sigma)$. Validation uses independent calibration and validation model sets (30 each) on five datasets.

Derived instances. Each of the eight instances defines an `ExplanationSystem` with domain-specific types and minimal (two-element) Rashomon witnesses. The attribution instance generalizes Bilodeau et al. [2024] from Shapley axioms to the abstract framework; the causal instance derives the Rashomon property from Markov equivalence [Verma and Pearl, 1991].

Empirical validation. Seven domains are validated with dose-response designs: underspecification is varied and instability measured, with negative controls confirming stability when Rashomon is absent. *Mathematics:* $Ax = b$ with four solvers, null-space dimension 1–10 (Mann–Whitney U). *Biology:* codon entropy across 120 cytochrome c sequences grouped by degeneracy (Kruskal–Wallis). *Gauge theory:* $\mathbb{Z}_2$ edge-label orbit variance on triangle graphs. *Linguistics:* four parsers on 50 ambiguous/50 unambiguous sentences (Wilcoxon). *Crystallography:* phase retrieval RMSD with/without positivity constraints (Mann–Whitney U). *Computer science:* census disaggregation KL divergence vs. county count (Spearman ρ). *Statistics:* causal discovery (PC, GES) on Asia network, 100 seeds (Mann–Whitney U). All tests use $\alpha = 0.05$, non-parametric statistics, and fixed seeds. No multiple-comparison correction across domains; binomial $p = 0.008$ provides combined assessment.

Reproducibility. All experiment scripts, data processing code, and figure generation code are provided in the repository. The complete experimental pipeline can be reproduced with `make validate` (∼5 min). The Lean formalization can be rebuilt with `lake build` (∼5 min). All paper versions can be compiled with `make paper`. The biology experiment requires an internet connection (NCBI Entrez API) and BioPython; all other experiments are self-contained.

Extended Data

Extended Data Table 1: Three ML instance instantiations

The following three ML explanation types are formal instances of the abstract framework, demonstrating that the impossibility is not merely analogical but structurally identical across explanation types.

Table 5: **Extended Data Table 1: ML instance instantiations.** Each row defines an explanation system $(\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ and identifies the Rashomon witness and resolution. All three are Lean-verified with zero domain-specific axioms (constructive variants use `Bool`/`Unit` types with Rashomon proved via `decide`).

Instance	$\mathbf{\Theta}$	$\mathcal{H}$	Y	**Witness**	**Resolution**
Feature attribution	Trained models (different seeds)	Feature rankings	Test predictions	Two GBDTs rank features j, k oppositely ($\rho = 0.9$)	DASH: ensemble average over M models
Attention maps	Fine-tuned transformers	Attention distributions	Classification output	Two DistilBERTs assign argmax attention to different tokens (19.9% flip rate)	Averaged attention rollout
Causal discovery	DAG structures	Edge orientations	Conditional independence structure	Chain $A \to B \to C$ and fork $A \leftarrow B \to C$ share CI structure	CPDAG: report undirected where orientation is ambiguous

Extended Data Figure 1: Universal η law

Data availability

Biology data: 120 eukaryotic cytochrome c coding sequences from NCBI RefSeq, retrieved via `CYCS[Gene] AND mRNA[Filter] AND refseq[Filter]`. Census data: U.S. Census Bureau 2020 Decennial Census county-level populations (variable P1_001N, downloaded via Census Bureau API). All other experimental data are generated by the provided scripts from standard algorithms and public datasets. Experiment results (JSON) are included in the repository.

Code availability

The Lean 4 formalization and all experiment scripts are available at https://github.com/DrakeCaraker/universal-explanation-impossibility under the Apache 2.0 licence.

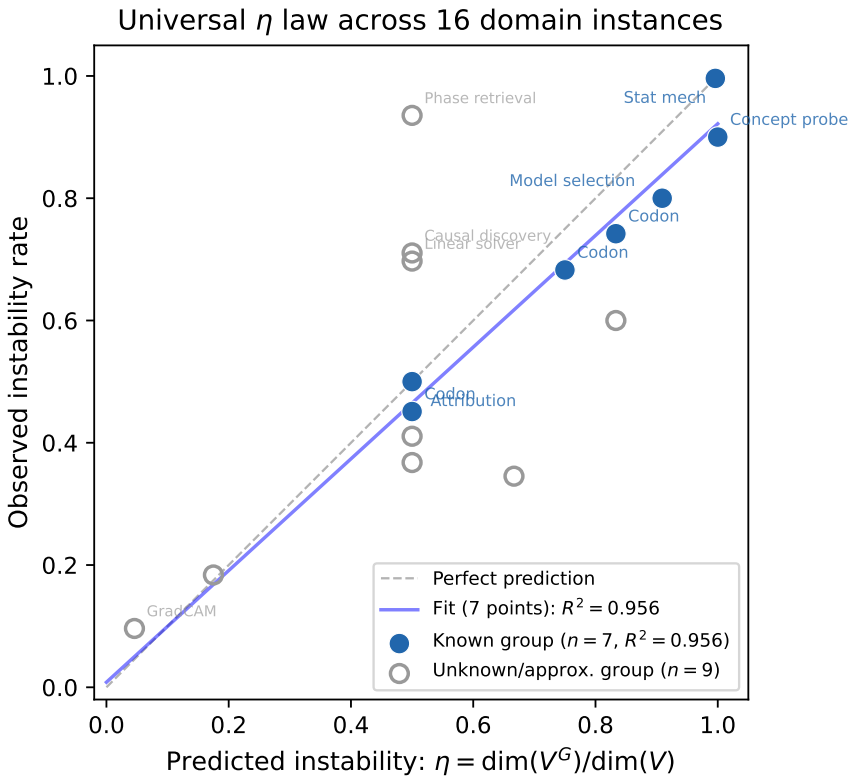

Figure 3: **Extended Data Figure 1: The universal η law.** The character-theoretic formula $\eta = \dim(V^G)/\dim(V)$ predicts observed instability across 16 domain instances. Filled blue circles: 7 instances with well-characterized symmetry groups ($R^2 = 0.957$, $p = 1.4 \times 10^{-4}$; leave-one-out $R^2 = 0.79$). Open grey circles: 9 instances with unknown or approximate groups ($R^2 = 0.25$). Dashed line: perfect prediction ($y = x$). The bottleneck is group identification, not the law itself.

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Competing interests

The authors declare no competing interests. D.C.'s employment at Oura Health is unrelated to this work.

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